Paper2PPT: AI-Driven Research Paper to Presentation Generator

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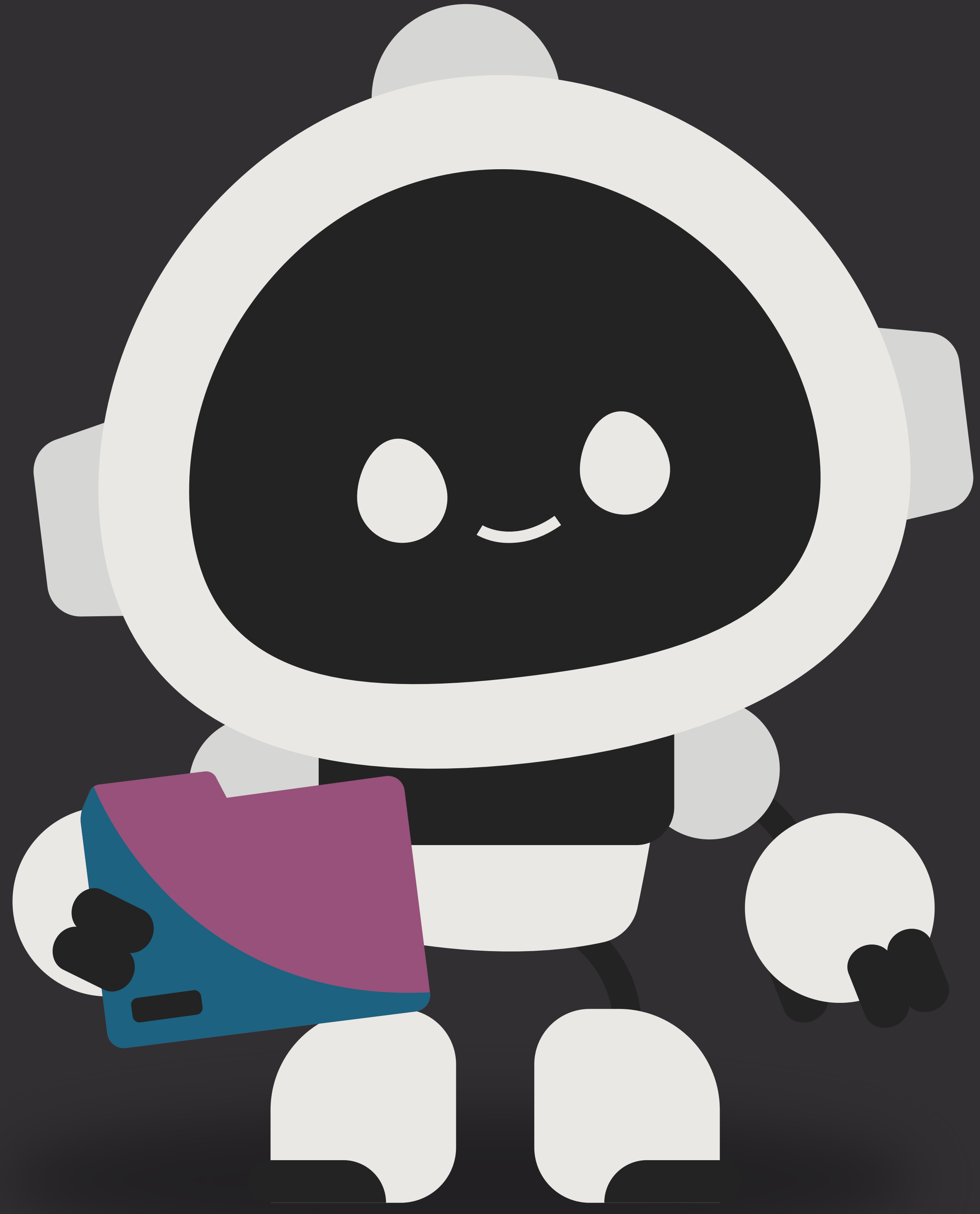
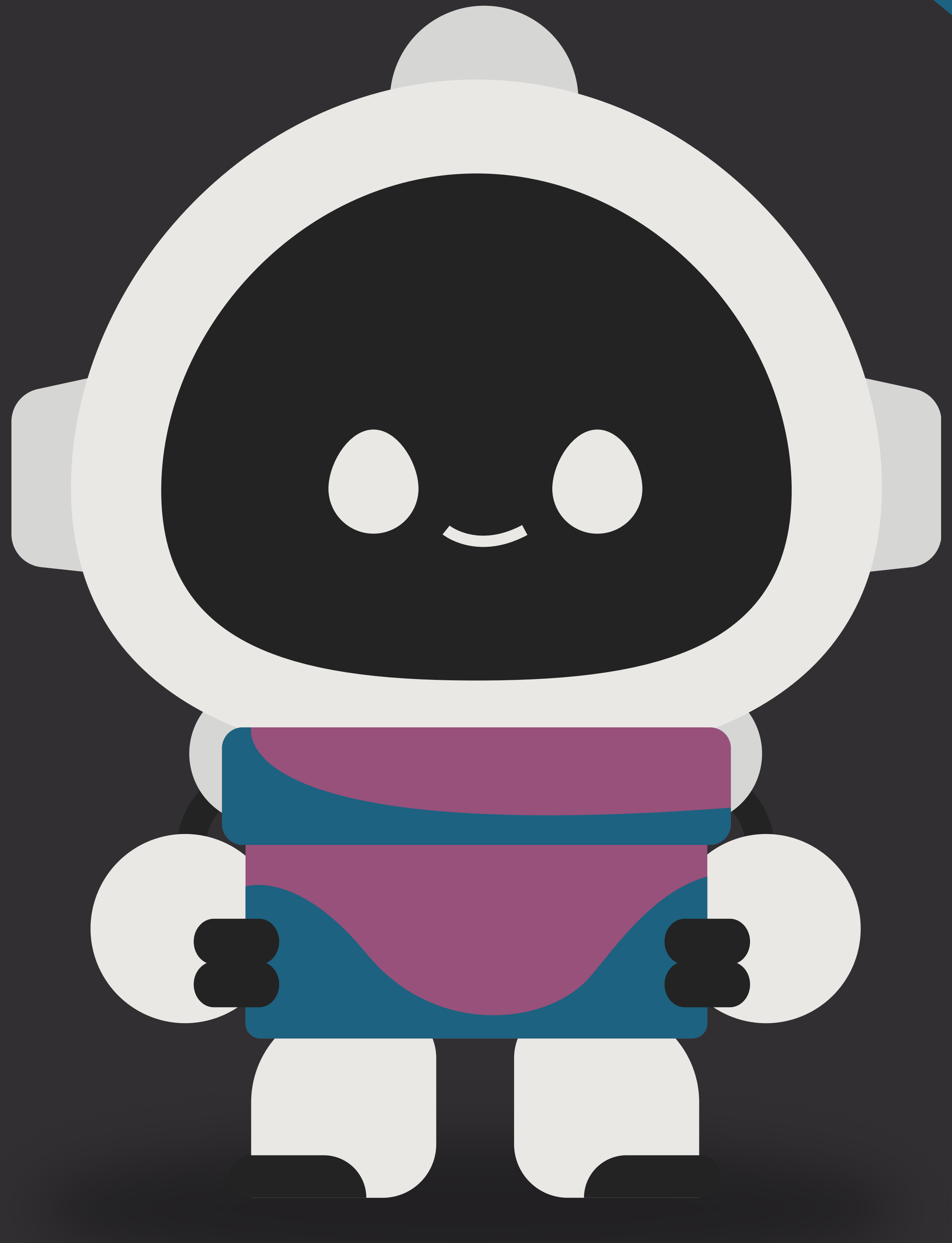
INTRODUCTION

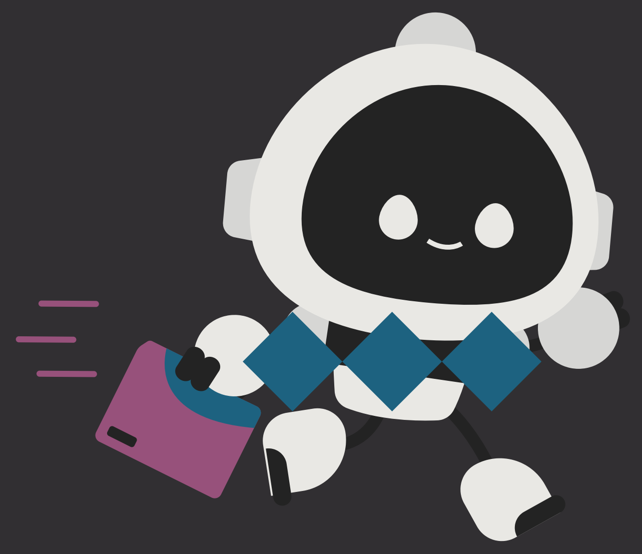
This project showcasing the practical use of Large Language Models (LLMs) in automating tasks.
Our system uses an Agentic AI approach to understand research papers and generate structured PowerPoint presentations autonomously.
It demonstrates how AI agents can plan, and act to simplify academic workflows efficiently.



TECH STACK

- LANGUAGE - PYTHON
- FRONTEND - STREAMLIT
- LLM - OLLAMA(3.2)
- API INTEGRATION - SEMANTIC SCHOLAR,CROSSREF,ARXIV

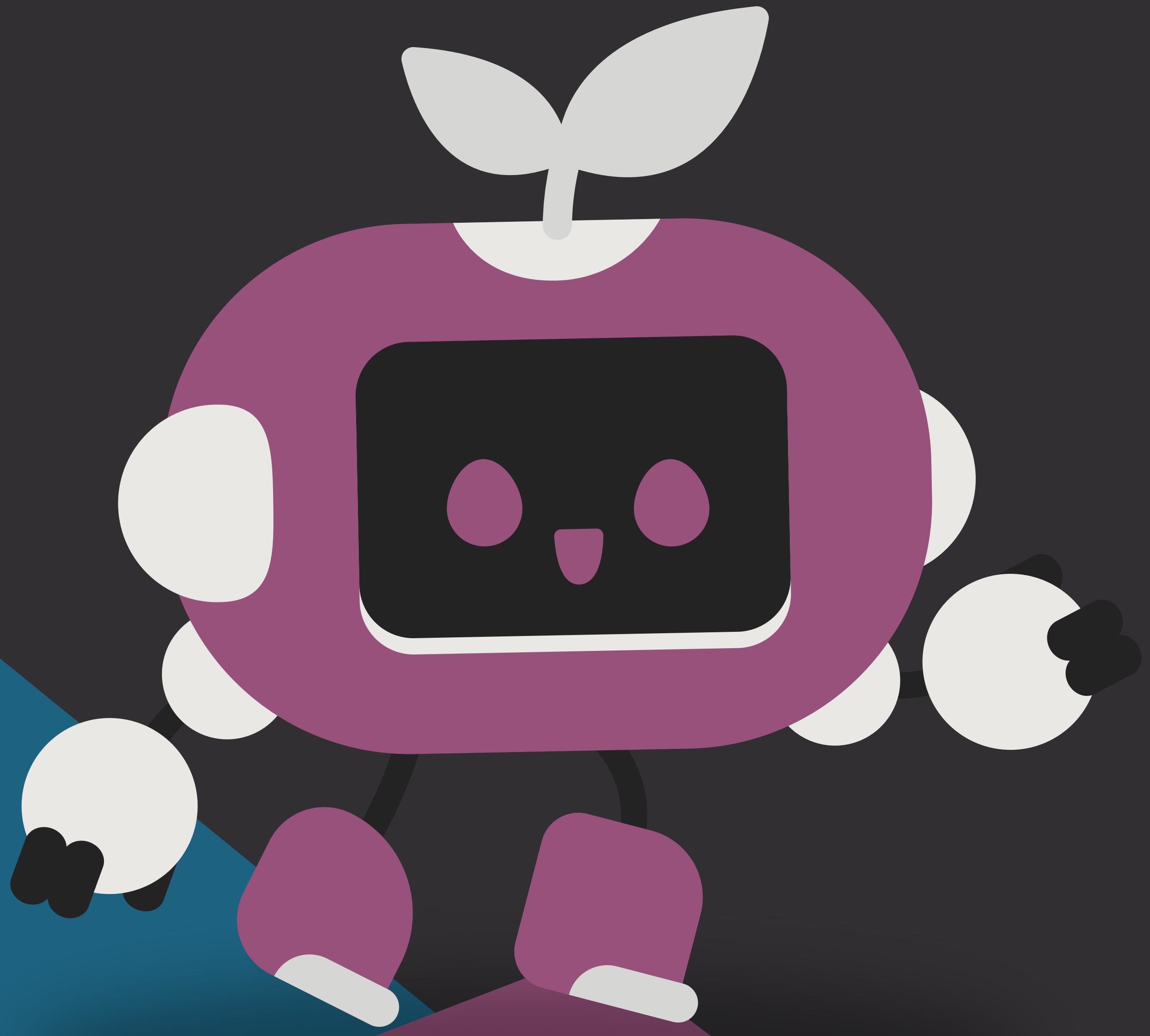




KEY FEATURES

API Integration: Fetches authentic research papers and abstracts instantly from (arXiv, Semantic Scholar, CrossRef)
Ollama (Llama 3.2) Model: Generates concise, context-aware slide content using local LLMs.
AI Safety Layer: Filters irrelevant or unsafe text while ensuring academic tone.
Smart PPT Builder: Automatically designs well-structured slides with colors, fonts, and AI visuals.
Streamlit UI: Provides smooth progress tracking, interactivity, and beautiful gradient themes.

SYSTEM ARCHITECTURE



- Frontend: Streamlit web interface with interactive components.
- Backend: Python engine integrating LLM (Ollama), Semantic Scholar API, and PowerPoint library.
- Steps:

1. Data Fetching (Paper Retrieval)
2. Text Summarization (LLM Reasoning)
3. Content Structuring (Slide Generation)
4. Presentation Design (PPT Creation)

METHODOLOGY

- User inputs a paper title, DOI, or link
- Paper metadata fetched via APIs (Semantic Scholar, arXiv, CrossRef)
- Content processed using LLM (Ollama 3.2)
- AI generates short, structured bullet points
- Slides built dynamically using Python-PPTX library

RESULTS

- Generates full PowerPoint presentations automatically from a single paper link or title.
- Slides maintain a consistent academic tone and professional visuals.
- Significantly reduces manual effort in preparing academic presentations.

CONCLUSION

This project applies Agentic AI concepts to automate research summarization using LLMs, turning complex papers into clear presentations. It highlights how intelligent agents can independently reason, generate, and organize information—showcasing the future of AI-driven academic tools.

THANK YOU